Initial Conditions at Labor Market Entry: Unequal Effects by Parental Income

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This paper examines the unequal impact of initial labor market conditions by parental income and their role in shaping long-term labor market outcomes. Using large Canadian administrative tax data, I explore the relationship between parental income, aggregate economic conditions, and wage dynamics for new labor market entrants. I find individuals experience significant and persistently lower earnings when entering the labor market during an economic downturns. These effects are unequal across the parental income distribution, with individuals from lower-income families facing larger earnings declines during recessions. A 1 p.p. increase in the initial unemployment rate reduces annual earnings by $2,000 real CAD for all workers over the first decade after graduation, with earnings declining 5 percentage points more for those from families in the bottom 20% of the income distribution. These disparities are largely mitigated when accounting for sector of employment, suggesting that sectoral selection plays a critical role. I develop a heterogeneous agent model incorporating financial constraints and frictional sector choice. The model highlights the role of parental income as a safety net: individuals from low-income families are closer to borrowing constraints and more likely to enter low potential earning sectors, especially in the presence of search frictions. Recessions amplify this mechanism by reducing job-finding rates, further limiting access to high-paying sectors. Counterfactual simulations suggest targeted policies, such as unemployment insurance for new labor market entrants, could mitigate scarring effects and improve sectoral mobility for disadvantaged workers.

1. Introduction

This paper examines the persistence of intergenerational income inequality, focusing on the interplay between parental income and initial labor market conditions. Parents can provide insurance against idiosyncratic income risks during their children's lifetimes, serving as a financial safety net for their adult children. For financially constrained individuals entering the labor market, this safety net can be particularly crucial to insulate against economic shocks.

At the same time, the conditions under which individuals enter the labor market have long-lasting effects. College students who graduate and enter the labor market in a recession experience average cumulative lifetime earnings losses of 5 percent and decreased wages for up to 10 years (Oreopoulos et al. 2012)). Existing research largely treats these initial labor market effects as uniform across individuals, overlooking heterogeneity by socioeconomic status. What is still not well understood is how the effects of initial labor market conditions differ across the parental income distribution. This paper addresses this gap by investigating how initial labor market conditions shape the long-term earnings and employment outcomes across the parental income distribution.

I introduce and establish evidence for a parental insurance mechanism, whereby individuals from lower-income families find themselves closer to the borrowing constraint and more likely to be liquidity constrained compared to their peers with better-off parents. These financial constraints exacerbate larger search frictions in the high potential earnings sectors, those with lower-income parents are more likely to sort into jobs with low potential earnings and human capital growth. When initial labor market conditions worsen, such as during recessions, this mechanism is further intensified as the hiring rate declines, amplifying the disparities in labor market outcomes across the parental income distribution.

To analyze this mechanism, this paper uses a rich dataset of Canadian administrative records that links tax files, post-secondary education records and parental income tax data. This dataset allows me to track individuals across their early careers with a high degree of precision and measure how initial labor market conditions affect earnings and employment outcomes across the parental income distribution. I find that initial labor market conditions affect entrants across the parental income distribution, but these effects are unequal. I document that when initial labor market conditions are worse and the initial unemployment rate is 1 percentage point higher, yearly earnings are on average 10% lower than trend, and these lowered earnings persist for 8 years after graduation.

One of this paper's main contributions is documenting the unequal effects by parental income. I find that labor earnings decline 5 percentage points more for entrants from the poorest 20% of families. Interestingly, when conditioning for broad labor market sector at the 2-digit NAICS level, differences by parental income groups are mitigated and there are no significant differences by parental income groups. This suggests that selection into employment sector plays an important role in explaining the variation in earnings responses to initial labor market conditions.

Using these empirical findings, I build a heterogeneous agent model with different initial assets and search frictions. What is key in my model is a frictional choice of labor market sector with different human capital accumulation and income processes. The model groups jobs into two labor market sectors. The first is a skill-complementary sector which has higher earnings potential and higher returns to human capital. The second is a skill insensitive sector with low potential earnings and where crucially earnings do not depend on human capital. This follows from Huckfeldt (2022) and more generally Lundqvist and Sargent (1998). I use the model to quantify the importance of initial assets and human capital, and asses their role in amplifying or mitigating initial conditions.

My primary contribution lies in being among the first to document disparities in earnings outcomes for graduates entering the labor market during a recession, based on parental income groups. These findings build on an empirical literature studying the effects of graduating in a recession. Additionally, I contribute by developing and solving a quantitative model, offering a theoretical framework to explain and isolate the economic mechanism which complements the predominantly empirical focus of existing research.

Related Literature: The literature finds graduating in a recession has a lasting negative impact on wages. Oreopoulos, von Wachter and Heisz (2012) document that a recession at labor market entry leads to lower labor earnings for up to the next 10 years. The authors examine male college graduates in Canada from 1982 to 1999, and find that graduating in a recession amounts to a decrease in about 8% of cumulative lifetime earnings. Kahn (2010) finds similar persistently lower wages in male college graduates in the US from 1979 to 1989 using the NLSY79. Similar patterns are found by Genda et al. (2010) and Speer (2015) among male high school graduates. The initial negative impact for high school graduates is larger than that for college graduates, and therefore the income losses are more substantial. However, little is known about the differential effects of labor market entry in a recession across the parental income distribution.

To the best of my knowledge, no paper studies the effects of parental income on graduating and entering the labor market in a recession. A growing empirical literature documents the persistence of intergenerational income inequality and the role of parental income in influencing children's lifetime earnings, including Chadwick and Solong (2002) and Chetty et. al. (2014) in the United States and Corak (2013) in Canada, as well as the role of post-secondary education (see for example Torche, 2011 and Chetty et. al., 2017). The intergenerational relationship of employers between parents and children has recently been documented empirically (Stinson and Wignall, 2014; Kaila et. al., 2024) and is increasing in paternal earnings (Corak and Piraino, 2011), which is expected to widen inequality by family income.

More broadly, family insurance has long been a topic of interest. The literature has demonstrated that parents do insure children by making transfers to less well-off children, as found by Cox (1990), McGarry (2016) and Ameriks et al. (2016). Choi, McGarry and Schoeni (2016) find that extended family income affects individuals' own consumption, and Boar (2021) shows that parents accumulate precautionary savings to insure their children against labor income risk. Moreover, Kaplan (2012) demonstrates moving into the parental home is an important channel to insure against labor market risk, focusing on the effects for male high-school graduates.

The paper is organized as follows. Section 2 describes the data and empirical results, section 3 presents the model, Section 4 details several counterfactual exercises and section 5 concludes.

2. Empirical Section

We begin by documenting several novel empirical facts. This section starts with a description of the administrative tax data and definition of parental income used throughout this paper. I then look at the empirical findings on the effects of earnings with parental income. I present aggregate means by cohort, followed by regression analysis of the microdata.

2.1. Data Description

I use Canadian administrative data that combines a panel of individual tax files linked to the tax files of family members, together with a database of post-secondary students in Canada (the T1FF-PSIS). My sample period focuses on students in the years 2006 to 2020. I use the annual unemployment rate at graduation to characterize the initial labor market conditions, following the literature in Oreopoulous et al. (2012). Figure 10 in the appendix shows the variation in the unemployment rate in Canada over time, with recession periods shaded in light gray. The main recession in my sample period is the Great Recession of 2008-09, during which the national unemployment rate for workers aged 15 and up increased by approximately 2 percentage points. In 2015, an oil shock dampened GDP and moderately increased the unemployment rate into 2016, though the economic contraction was not formally classified as a recession. I exclude graduates and new labor market entrants during the Covid recession in 2020 due to the lack of available data in the years following the recession.

2.2. Measure of Parental Income

This paper measures parental income by using the mean parental pre-tax income for each individual observed in the dataset. All income variables are inflation adjusted. This measure of parental income therefore incorporates not only parental labor income, but also income including capital gains, returns on interest and business income, making it a more holistic measure of total parental income and overall wealth. Taking the average parental income over several years also helps ensure we are measuring a good proxy of permanent parental income that is less influenced by temporary income fluctuations. Throughout the paper, I show results by parental income classified into two groups, those in the bottom 20% and top 80% of average real parental income, where the quintile categories are CPI adjusted and defined across all years. My analysis will compare individuals with different groups of parental income who have the same graduation year. I follow the outcomes of these individuals over years after graduation. The intergenerational income inequality literature commonly analyses parental income in quintiles, with specific interest on individuals from the bottom income quintile. The bottom 20% group also roughly corresponds to poor and liquidity constrained individuals, who Kaplan and Violante (2014) refer to as the poor hand-to-mouth. In robustness checks, I also run my analysis using groups defined on mean parental income before graduation, five quinitle groups as well as using continuous parental income. The results are largely similar.

2.3. Cohort Level Results

I begin by presenting aggregate labour outcomes for graduation cohorts, analyzing the relationship between parental income and the unemployment rate (UR) at the time of graduation. Figure 1 below shows average labor earnings by group in real Canadian dollars, on the y-axis, plotted against years after graduation on the x-axis. This figure presents raw group means by graduation cohort, parental income group, and years after graduation without conditioning on covariates and includes individuals with zero earnings. The year of graduation is denoted at time zero with a dashed line. Each subplot corresponds to a parental income group. In the figure below, we can see individuals whose parents are in the bottom 20% of parental pre-tax income in the left subplot, and individuals whose parents are in the top 80% of the parental income distribution in the right subplot. Each line denotes a graduating cohort that is colored according to the annual total unemployment rate at graduation, with light blue representing graduating into a good labor market with low unemployment, and orange denoting a more difficult initial labor market with a high unemployment rate.

We observe persistent and long-lasting differences in labor earnings by initial aggregate conditions. For all levels of parental income, Figure 1 shows that individuals who graduate and enter the labor market in better initial conditions, in blue, have higher earnings on average for approximately eight to ten years compared to those who graduate into a worse labor market. This finding is in line with Oreopoulos et. al (2012), who document lasting scarring effects for male college graduates in the 1980s and 1990s. The persistence of initial labor market conditions is surprising, as the effects on mean wages far outlast the duration of the recession, which ended in 2009, and wage effects remain even after elevated unemployment rates subside by 2013.

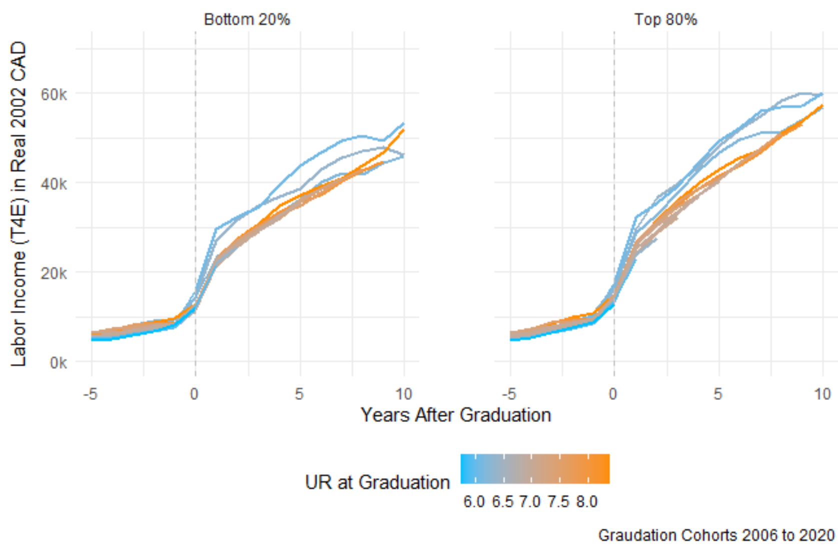

FIGURE 1. Labor Earnings by Parental Income Group and Initial Labor Market Conditions

Moreover, we can also see differential effects of earnings across the parental income distribution. In Figure 1, we observe that labour earnings have different profiles by parental income groups irrespective of initial labour market conditions, with individuals from low-income backgrounds having both a lower level and a flatter slope of earnings on average. This effect is not immediate, since Figure 1 visualizes only labour earnings, which excludes potential transfers from richer parents to their children. It is therefore not obvious how labour earnings would be directly affected by parental income group. I take these average differences into account in the micro level regressions in the following section.

2.4. Regression Specification

To further understand the relationship between initial labor market conditions and parental income, I analyze the mircrodata via regression analysis. I focus on individuals with at least three years of non-missing parental income to try to proxy for permanent income and, to the best of my ability, and reduce effects of temporary parental income fluctuations. I concentrate on individuals who are aged 26 and younger at the time of their first graduation as they are less likely to have extensive work experience prior to college. My sample contains approximately 200 thousand unique individuals and 1.2 million individuals over time who graduated from post-secondary in the sample period. Post-secondary education is defined as education following secondary school, and therefore after high school. It includes individuals who complete one to two year associate degrees and certificates at colleges, as well as those who finish bachelor's degrees and advanced degrees including masters' degrees.

I run the following regression specification dynamically over years after graduation, denoted by $j = t - c$, where c is the graduation year and t is the calendar year:

$$
\begin{aligned}
y_{ctpi} = \alpha + & \underbrace{\sum_{j=0}^{10} \beta_{p,j=(t-c)} \text{Parental Inc Group} \times \mathbb{1}_{j=t-c}}_{\text{Average effect of parental income group on indiv earnings}} \\
& + \underbrace{\sum_{j=0}^{10} \gamma_{0,j=(t-c)} UR_{c0} \times \mathbb{1}_{j=t-c}}_{\text{Average effect of initial UR rate}} + X_i' \delta + \phi_t \\
& + \underbrace{\sum_{j=0}^{10} \gamma_{p,j=(t-c)} UR_{c0} \times \text{Parental Inc Group} \times \mathbb{1}_{j=t-c}}_{\text{Additional effect of having parental income group } p} + u_{ctpi}
\end{aligned}
\tag{1}
$$

In my main specification, the dependent variable y_{ctpi} denotes the individual i's real labor earnings in CPI adjusted Canadian dollars, including zeros if the individual is has no labor earnings. I denote the graduation cohort year by c, calendar year t and parental income group p. In some specifications, the dependent variable is restricted to positive labor earnings, and I can therefore look at log earnings. I run further robustness with positive labor earnings above ten thousand 2002 real Canadian dollars, which approximates one year of full-time minimum wage earnings throughout my sample. The effect over time is denoted by $j = (t - c)$, the number of years after graduation.

My main variables of interest are the mean effect of parental income on earnings over time, captured by $\beta_{p,j}$, and the average effect of the initial unemployment rate on earnings in the years following graduation by parental income group p. The average effect of the initial unemployment rate at graduation UR_{c0}, is denoted by $\gamma_{0,j}$ and captures the effect for the reference parental income group, which is the group of parents in the bottom 20% of the parental income distribution. The interaction term $\gamma_{p,j}$ is the additional affect of the initial unemployment rate at graduation for individuals whose parents are in the other parental income groups in the top 80% of the parental income distribution. This interaction term $\gamma_{p,j}$ is a key parameter of interest; if it is significantly different than zero, then there is evidence of that initial labor market conditions affect entrants differently depending on parental income. With the reference group being those from the lowest socio-economic background, a negative interaction term $\gamma_{p,j} < 0$ signifies that individuals with richer parents have lower earnings when the unemployment rate at graduation UR_{c0} is one percentage point higher. Conversely, if the interaction term is positive, $\gamma_{p,j} > 0$, individuals from poorer family backgrounds fare worse and have lower earnings than their peers from richer backgrounds when the initial unemployment rate increases.

The regression specification conditions on a number of co-variates. In the baseline specification, I condition on the individual's current age as well as year fixed effects. I progressively condition on additional covariates including gender, currently being enrolled in post-secondary, the degree type, and broad major grouping. I also use information on the individual's current principal industry of employment at the two-digit NAICS level. Degree type is classified by the International Standard Classification of Education (ISCED), which captures differences in post-secondary degrees including between a two-year degree, bachelors' degree or equivalent, masters' degree etc. Major groupings in the PSIS has several available variables categorizing field of study by the Classification of Instructional Programs (CIP) Canada 2016 version, including a primary grouping code (CIPPG), a 2-digit code, and a 4-digit code. I condition on the primary grouping code (CIPPG) in the main results, and have run robustness checks using more detailed 2-digit and 4-digit CIP major codes.

2.5. Regression Results

I find that individuals who graduate with a 1 percentage point higher unemployment rate have persistent earnings declines of approximately $1700 real 2002 CAD per year, equivalent to nearly $2700 in 2024 Canadian dollars. This result is robust to numerous control variables including age, gender, degree, type of institution, and major grouping. Moreover, the effect of initial labor market conditions on the path of earnings is highly persistent, with earnings being on average $1700 CAD lower per year for 10 years after graduation, when the unemployment rate at graduation is 1 percentage point higher. That is to say, when comparing two individuals with the same set of observables who are in the same parental income group, those who graduate from post-secondary in a year with a worse labor market and higher unemployment rate earn on average CAD $1700 less labor earnings per year for ten years. This effect is therefore economically large.

Interestingly, the impact of initial conditions on earnings trajectories is heterogenous by parental income group, which is a novel finding and one of this paper's main contributions. Those in the bottom 20% of the parental income distribution have larger earnings declines compared to when the unemployment rate at graduation is 1 p.p. higher, with their earnings declining by on average $2200 real CAD per year for the first 3 years after graduation. Meanwhile, for individuals with parental income in the top 80% of the parental income distribution, labor earnings decline by on average only $1500 real CAD per year for the first three years after graduation. Thereafter, both groups face similar earnings declines when the initial unemployment rate is lower, regardless of their ranking in the parental income distribution as shown in Table 3 in the Appendix.

Thus, individuals from the lowest socio-economic class are more vulnerable to initial labor market conditions, namely the unemployment rate at graduation, compared to their peers from middle and upper class backgrounds. This relationship is not purely mechanical, since I am measuring the response to labour earnings that do not include direct transfers from parents to children. My findings hold even when conditioning on observables including gender, degree type, and major grouping. I use additional robustness checks to condition for the top 3 and top 10 institutions as well as living at the parental home, and the results are similar.

Notably, when conditioning on employment industry at the two-digit NAICS level, the different responses in earnings by parental income group are no longer statistically significant. This suggests that industry selection plays an important role in explaining the variation in earnings responses to initial labor market conditions. I explore the role of industry in greater detail in Section 2.6. These findings are consistent with parents acting as a financial safety net. Individuals from higher and middle income families are further from the borrowing constraint, and are more able to spend time selecting jobs with higher earnings potential in the first three years after graduation. In comparison, those individuals from low income families lack the liquidity to do so, and are more pressed to take whichever job is available as they are closer to the borrowing constraint. During an economic downturn, particularly the 2008-09 Great Recession, employment in high potential earnings sectors becomes harder to find, further amplifying the impact of liquidity constraints and labor market conditions on industry selection and widening differences by parental income group.

The average earnings of both groups eventually recover to their long run trend. Those from the bottom 20% of the parental income distribution recover slightly faster in approximately nine years after graduation, while those in the top 80% of the parental income distribution take on average ten to eleven years after graduation to return to the long-run trend. This is in line with the observation that those from the bottom 20% of the parental income distribution have a lower average trend of labor earnings compared with individuals from wealthier parents, and therefore their average earnings catch up slightly faster.

Looking at individuals with positive labor earnings only who are in the workforce, we can analyze the effect of initial labour market conditions on log earnings, $\log y_{ctpi}$. Table 4 in the Appendix documents the results. When conditioning on individual positive earnings, individuals with worse initial labor market conditions likewise have a significant and persistent decrease in their earnings. On average, I find that individuals who graduate and enter the labor market with a 1 percentage point higher unemployment rate have 3 percent lower earnings each year for six years following graduation. This persistent result is robust to controls for age, gender, being in school, degree and institution type as well as major and year fixed effects.

Once again, initial labor market conditions have a different impact on earnings by parental income group. Similarly to the results shown in levels, individuals in the bottom 20% of the parental income distribution are more affected by the initial unemployment rate compared to their peers in the top 80%. This heterogeneous effect is concentrated in the first two years after graduation, after which both groups experience similar declines in earnings. Workers from the poorest 20% families have on average approximately 5.5% lower earnings in the year of graduation, and 3.5% lower earnings one year after graduation, all else equal. For workers from the middle and top of the family income distribution, earnings decrease less relative to their trend, with earnings falling on average only 2.5% in the graduation year and 1.5% in the year after graduation when conditioning for the same observables including age, gender, degree type, major group and two digit NAICS industry of employment.

To approximate full-time workers, I focus on individuals whose annual earnings exceed the real full-time minimum wage, which is approximately CAD $10,000 in 2002 constant dollars. Consistent with earlier findings, earnings for these proxied full-time workers are significantly influenced by initial labor market conditions and vary across parental income groups. The results are shown in Table 5. A one percentage point increase in the unemployment rate at the time of graduation is associated with an average annual decline in labor earnings of 2% per year over the first seven years post-graduation, holding other factors constant. These effects are both highly persistent and economically significant, underscoring the substantial scarring impact of adverse labor market conditions at graduation on long-term earnings.

The effects on earnings vary significantly by parental income group. Individuals from the bottom quintile of the parental income distribution experience larger earnings declines relative to trend compared to their peers from middle- and upper-income backgrounds. Earnings for those in the bottom 20% of the parental income distribution decline by approximately 3.5%, while the corresponding decline for those in the top 80% is only 1.5%. These differences remain robust when controlling for observable characteristics as shown in Table 5, including gender, degree, major, age, and broad industry grouping at the two-digit NAICS level, ensuring comparability across individuals. Notably, this differential effect persists for up to three years after graduation, after which both groups experience similarly lowered earnings for seven years after graduation when entering a worse initial labor market.

Together these results underscore that individuals from different parental income groups experience different effects of initial labor market conditions. The early post-graduation years, which are critical for career development, reveal significantly different responses to the initial unemployment rate by parental income group, particularly during the first three years following graduation. Workers from low-income families are particularly vulnerable to adverse initial labor market conditions, as they often lack a parental safety net and are more likely to face borrowing constraints. Consequently, they are more inclined to accept available employment out of financial necessity, and are more likely to sort into low potential earnings industries, despite having similar observables to their peers from more affluent parents. The following section explores the role of employment sector in shaping these outcomes.

2.6. Labor Market Sector

As seen in the micro-regressions described above, industry of employment plays a significant role in accounting for differences in wage trajectories by parental income group. I delve into this further and classify sub-industries into two groups to better understand the role of industry sector. To do so, I use an auxiliary sample of individual tax data from the LAD, which is a one percent sample of Canadian tax payers of all ages. This sample is a panel which includes individuals in my main post-secondary sample as well as many older individuals and individuals who may not have attended post-secondary. I use this sample, spanning from the year 2000 — when NAICS codes were introduced — through 2016, to classify 3-digit NAICS subindustries into high- and low-earnings potential groups based on average wages during prime earning years between ages 45 and 55. Earnings for individuals between the ages of 45 and 55 are a good proxy for earning potential, reflecting accumulated work experience and close to peak earnings before retirement.

I use machine learning techniques, namely k-means clustering, to group industry subsectors into an optimal number of clusters. Using mean and median earnings for individuals aged 45-55, I find the data is optimally classified by two clusters that distinguish subindustries with high earnings potential and low earnings potential. The classifications, along with various robustness checks, are shown in the Appendix.

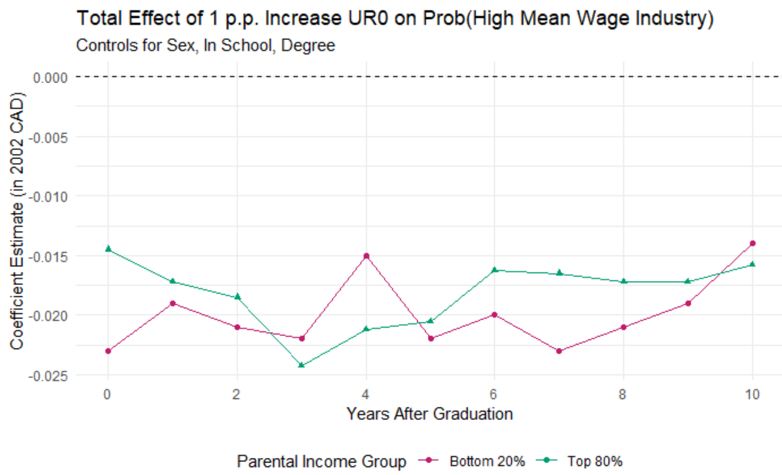

FIGURE 2. Probability of employment in sector with high earnings potential by initial UR

I run a similar regression to equation 1 on post-secondary graduates, with the dependent variable being the probability of working in an high earnings potential industry. Once again, I condition on numerous observables including age, gender, current enrollment in post-secondary, institution type, degree and major. When the unemployment rate at graduation increases by 1 percentage point, I find that the probability of working in a high earnings potential subsector decreases across parental income distribution.

Individuals who face worse initial labor market conditions are significantly less likely to work in a high earnings potential industry. These effects are true across the parental income distribution. With an increase in the unemployment rate at graduation by one percentage point, new labor market entrants are approximately 2 percentage points less likely to work in a high earnings potential sector, all else equal.

Additionally, these effects are highly persistent, lasting for a decade after graduation. Ten years after graduating from post-secondary, the probability of working in a subsector with high potential earnings remains approximately 1.5 percentage points lower compared to individuals who graduated and entered the labor market under better initial conditions. These effects are economically large, and in line with my previous findings documenting a significant reduction in earnings associated with worse initial labor market conditions. I also find that a higher initial UR rate is not associated with increased industry switching. From two years after graduation onward, there is no significant increase in job and industry switching associated with a higher initial unemployment rate. As a result, initial selection into industries is highly relevant and persistent, as initial sorting into sectors with low earnings potential is not offset by transitions into sectors with high potential earnings over time.

There are slight differences by parental income group immediately after graduation, with lower income individuals being even less likely to start in a high earnings industry. However, these differences close after one of year of graduation. Initially, individuals from the bottom 20% of parental income distribution are 2.25 percentage points less likely to be in a high earnings potential subindustry compared to better initial labour market conditions. For children in the top 80% of the parental distribution, they are still less likely to be in industries with high earnings potential, but that effect is notably smaller in magnitude at only 1.5 percentage points less likely. Differences by parental income converge a few years after graduation, and all individuals that graduate with a one percentage point higher rate of unemployment are less likely to be in subsectors with high potential earnings.

Next, I condition on broad sector, such as construction or healthcare, to understand whether individuals are more likely to enter high potential earnings subsectors *within a broad sector* depending on initial labor market conditions. I find that that initial labor market conditions do affect the likelihood of working in a high paying subindustry, and they do so differently by parental income group. When comparing workers in different subindustries but working within the same broad two-digit NAICS industry, the probability of being in a high earning potential sector decreases for bottom 20% for two years following graduation, but is unaffected for middle and upper class individuals.

In the first two years after graduation , individuals with parents in the top 80% of the income distribution are not significantly affected by a change in the initial UR at graduation. However, those with parents in the bottom 20% of the income distribution are approximately 0.5 percentage points less likely to work in a high potential earnings subindustry. In the medium run, 4 to 8 years after graduation, both groups are somewhat more likely to work in a high earnings potential subindustry. However the effect is larger for those with better-off parents, who are 1 p.p. more likely to be in a high potential earnings sector, compared to only 0.5 pp for those from the families in the bottom 20 percent of the parental income distribution. In the years following, differences by parental income in the probability of working in a sector with high potential earnings converge.

Together these empirical results reinforce the importance of industry selection in explaining earnings differences across groups. In the following section, I develop a structural model to better understand the relationship between sector choice and earnings dynamics. The model will assist in rationalizing how sectoral choices take place in a context with search frictions and financial constraints, and help to isolate the mechanism underlying the earning dynamics observed in the data.

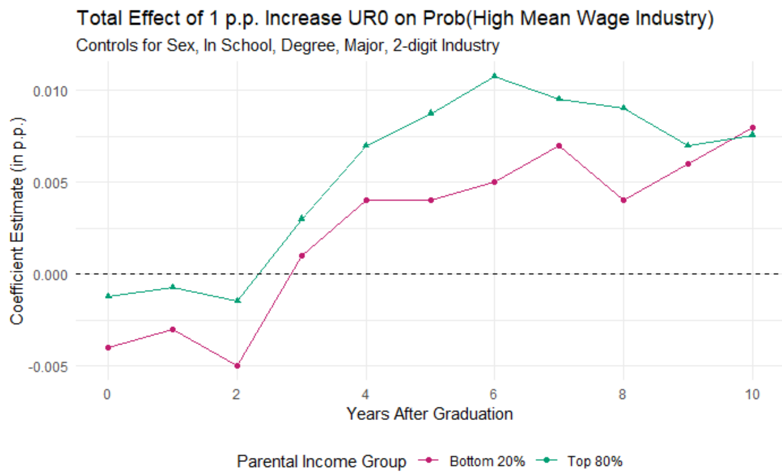

FIGURE 3. Probability of employment in high earnings sector by Initial UR, conditional on broad industry grouping

3. Model

Section 3 lays out the structure of the model. The model departs from a heterogeneous agent model with incomplete markets, where agents have different initial conditions and face a frictional labor market as well as borrowing constraints. I model parental income as a "safety-net" for new labor market entrants, where parental income determines initial assets a_0^p. Individuals face borrowing constraints which increases the reliance on initial parental wealth. Individuals are endowed with human capital level $h \in \mathcal{H}$, that will affect their earnings and employment outcomes. Every period, there is a frictional choice of labor market sector. Agents choose the labor market sector j or s among unemployment (U), low sector (L) and high sector (H). The sectors will differ in terms of earnings and human capital accumulation, as described below. In addition, agents choose the optimal levels of consumption c, savings a'. The state variables are current assets a, human capital h, current labor market sector j and the aggregate state of the economy $\mathcal{Z}$.

3.1. Primitives

Agents have CRRA utility with relative risk aversion γ

$$U(c) = \frac{1}{1-\gamma} c^{1-\gamma} \qquad \gamma > 1, \gamma \neq 1 \tag{2}$$

where c denotes the consumption level.

The individual labor income processes $y^j(h, \mathcal{Z})$ differ by labor market sector j or s, as well as human capital level h and the aggregate state of the economy $\mathcal{Z}$, similarly to Huckfeldt

(2022). Labor supply is assumed to be inelastic.

There are three labor market sectors: unemployment (U), the low-paying and low human capital growth sector (L), and the high-paying, high human capital growth sector (H). The labor earnings for each sector are as follows, where $h \geq 1$ and $0 < \alpha \leq 1$.

$$\text{Skill-complementary sector, H} \qquad y^H = \mathcal{Z} \times h^\alpha \tag{3}$$

$$\text{Skill-insensitive sector, L} \qquad y^L = \mathcal{Z} \tag{4}$$

$$\text{Unemployed, U} \qquad y^U = b^U \tag{5}$$

In the skill-complementary sector, denoted hereafter by the H sector superscript, earnings depend on both individual human capital h as well as the aggregate economic conditions $\mathcal{Z}$. The parameter α captures the decreasing returns of human capital to earnings. In the skill-insensitive sector, denoted henceforth by the L sector superscript, earnings do not depend individual human capital h, but only on the aggregate economic conditions $\mathcal{Z}$. Finally, if an individual is in unemployment, denoted by the U superscript, they receive an unemployment benefit b^U.

Human capital h evolves stochastically and heterogeneously in each sector, following Huckfeldt (2022). With probability π_s where s denotes the current labor market sector, next period human capital increases or decreases from its current value by one step on the human capital grid $\mathcal{H}$. With probability $(1 - \pi_s)$, next period's human capital remains the same as the current value of human capital. The transition probabilities vary by sector, as follows.

- Skill-complementary sector: $h' = \begin{cases} h + \Delta_{\mathcal{H}}, & \text{with prob } \pi_H \\ h & \text{with prob } 1 - \pi_H \end{cases}$
- Skill-insensitive sector: $h' = \begin{cases} h - \Delta_{\mathcal{H}}, & \text{with prob } \pi_L \\ h & \text{with prob } 1 - \pi_L \end{cases}$
- Not employed: $h' = \begin{cases} h - \Delta_{\mathcal{H}}, & \text{with prob } \pi_U \\ h & \text{with prob } 1 - \pi_U \end{cases}$

In the skill-complementary sector, human capital next period increases by one step with probability π_H, and with probability $(1 - \pi_H)$, human capital next period stays the same as current human capital. For the skill-insensitive sector, where earnings do not depend on human capital, there is a probability π_L that human capital depreciates next period, and

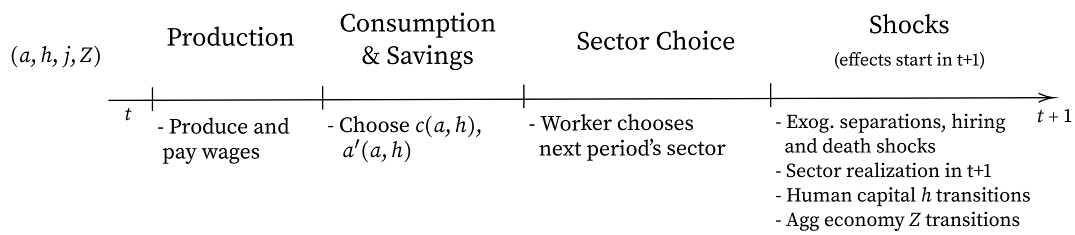

FIGURE 4. Timing of Model Each Period

otherwise with probability $(1 - \pi_L)$, human capital next period is equal to current human capital. The skill depreciation can be thought of intuitively as coming from forgetting skills that are not used while working in a menial job. When the individual agent is unemployment, human capital has the probability of depreciating to a lower level with probability π_U. In the calibration π_U will be larger than that of the skill-insensitive sector π_L. With remaining probability $(1 - \pi_U)$, the human capital of an agent in non/unemployment is unchanged in the following period.

The agents begin their life in the state of unemployment, with initial assets a_0^p determined by their parental income. For the next period, individuals can choose which sector to search in. They can either enter the skill-insensitive sector with probability one, or else attempt to be hired in the skill-complementary sector with hiring probability $p^H(h, \mathcal{Z})$ that is increasing in human capital h and in better economic conditions $\mathcal{Z}$. If the agent chooses to search in the skill-complementary sector but is not hired, that occurs with probability $(1 - p^H(h, \mathcal{Z}))$, they find themselves in unemployment again next period. The model abstracts from job to job transitions.

3.2. Model Timing

The timing of the model is as follows. Within a period t, an agent with assets a, human capital h, in labor market sector j and with aggregate economic conditions $\mathcal{Z}$ will first receive labor income $y^j(h, \mathcal{Z})$ as described in the income process section [NUMBER HERE] above. Next, the individual agent chooses optimal levels of consumption $c(a, h, j, \mathcal{Z})$ and savings $a'(a, h, j, \mathcal{Z})$. Following that, the individual chooses next period's sector j'. Finally, transitions and shocks occur whose effects are realized at the beginning of the next period.

Hiring and separation shocks can disrupt the next period realization of the chosen labor market sector. When the individual is in unemployment and chooses to apply in the skill-complementary sector, the hiring probability $p^H(h, \mathcal{Z})$ determines if they will be hired in the skill-complementary sector next period. Should the individual not be hired, they find themselves in unemployment once again following period.

Moreover, employed individuals in sector j face an exogenous separation shock η_j every period. When the separation shock hits, the individual will be in unemployment next period instead of their optimally chosen sector j'.

Human capital evolves as in the equation above, depending on the agent's sector next period. If no hiring or separation shocks occur, individual human capital evolves according to the individual's optimal sector choice for next period, j'. Similarly, the aggregate state of the economy $\mathcal{Z}$ transitions.

Agents face a constant probability of dying σ every period, as in the Blanchard (1985) model of perpetual youth. When an agent dies, at the beginning of the next period they redraw human capital from an initial distribution and a new individual is born in unemployment. I assume that assets are inherited from the deceased ancestor, which is a reduced-form way of inheriting all assets from the parents without explicitly modeling the parents' problem. This assumption also abstracts from bequest motives of parents to save for their children.

3.3. Equilibrium

The agent's problem in equilibrium is as follows. As outlined above, each agent chooses consumption c and savings a', and chooses to work in a sector $s \in \{U, L, H\}$. The aggregate economic conditions are $\mathcal{Z}$. Individuals receive labor income y which depends on human capital h, sector s, and the aggregate economy $\mathcal{Z}$. Individuals also receive a "safety-net" of initial income from parents a_0^p, and face a no-borrowing constraint.

The value function in unemployment is the following:

$$V^U(a, h, \mathcal{Z}) = \max_{c,a'} u(c) + \beta\mathbb{E}[(1-\sigma)\max\left\{V^L; p^H(h', \mathcal{Z}')V^H + (1 - p^H(h', \mathcal{Z}'))V^U\right\}] \quad (6)$$

$$\begin{aligned} \text{s.t.} \quad & c + a' = Ra + b^U \\ & a' \geq 0 \\ & h' = \begin{cases} h - \Delta_{\mathcal{H}} & \text{with prob } \pi_U \\ h & \text{with prob } 1 - \pi_U \end{cases} \end{aligned}$$

The individual agent chooses consumption and savings to maximize current utility as well as the expected utility of next period. Every period, there is a probability σ that the agent dies at the end of the period, in which case a new agent will be born in unemployment next period, inheriting assets but redrawing human capital. The process is described in more detail below. With probability $(1-\sigma)$ the agent does not die (lives on?) next period, and chooses to apply to the labor market sector $s' \in \{L, H\}$ which maximizes their present discounted value. This choice is denoted in the max operator. More specifically, the individual chooses between applying for the low sector (L) with hiring probability one , or applying for the high sector (H) with hiring probability $p^H(h', \mathcal{Z}')$. If the individual applies to the high sector but is not hired, with probability $(1 - p^H(h', \mathcal{Z}'))$, they remain in unemployment next period. The budget constraint is standard with income in unemployment equal to b^U and the human capital process described in section 3.1 above. I assume a no-borrowing constraint - this can be relaxed to an exogenous borrowing limit Φ without loss of generality.

In the skill insensitive sector (L), the value function is the following. Income and human capital are as described in section 3.1 above.

$$V^L(a, h, \mathcal{Z}) = \max_{c,a'} u(c) + \beta\mathbb{E}[(1-\sigma)(\eta V^U + (1-\eta)\max\{V^L; V^U\})] \tag{7}$$

$$\begin{aligned} \text{s.t.} \quad & c + a' = Ra + \mathcal{Z} \\ & a' \geq 0 \\ & h' = \begin{cases} h - \Delta_{\mathcal{H}} & \text{with prob } \pi_L \\ h & \text{with prob } 1 - \pi_L \end{cases} \end{aligned}$$

An individual in the skill-insensitive sector (L) who does not die next period is with probability η exogenously separated from their job into unemployment next period. If the individual is not separated, they choose between continuing in the low sector next period, or going into unemployment with the possibility of moving to the high sector thereafter.

Finally, the value function of an individual presently in the skill sensitive sector is shown below in equation (8).

$$V^H(a, h, \mathcal{Z}) = \max_{c,a'} u(c) + \beta\mathbb{E}[(1-\sigma)(\eta V^U + (1-\eta)\max\{V^H; V^U\})] \tag{8}$$

$$\begin{aligned} \text{s.t.} \quad & c + a' = Ra + \mathcal{Z}h^{\alpha} \\ & a' \geq 0 \\ & h' = \begin{cases} h + \Delta_{\mathcal{H}} & \text{with prob } \pi_H \\ h & \text{with prob } 1 - \pi_H \end{cases} \end{aligned}$$

Similarly to the value function in the skill-insensitive sector, an agent who survives into next period is, with probability η, exogenously separated from their job and will go into unemployment next period. Should the individual not be separated, the model allows them to choose to either continue in the high sector next period, or to go into unemployment next period with the possibility of switching sectors afterwards, as denoted by the choice inside the max operator.

3.3.1. Equilibrium Definition

An equilibrium is a set of value functions $V^s(a, h, \mathcal{Z})$ for each sector $s \in \{H, L, U\}$; a set of policy functions for consumption $c(a, h, s, \mathcal{Z})$, savings $a'(a, h, s, \mathcal{Z})$, and sector choice next period $s'(a, h, s, \mathcal{Z})$ given initial asset, human capital and sector distributions, a_0, h_0 and s_0 respectively ; and a stationary distribution of agents over individual and aggregate states $(a, h, s, \mathcal{Z})$.

The laws of motion implied by the value and policy functions across each state are as follows.

Let $\mu_L(h, a)$ be the mass of individuals with human capital level h and asset level a in the low sector with skill insensitive earnings. Let $\mu_H(h, a)$ be the mass of individuals with human capital level h and asset level a in the high sector with skill-sensitive earnings, and $\mu_U(h, a)$ be the mass of individuals with human capital level h and asset level a in the unemployment sector.

$$E_{UH} = p^H(h, \mathcal{Z})V_H^{'} + (1 - p^H(h, \mathcal{Z}))V_U^{'} \tag{9}$$

3.4. Calibration

TABLE 1. Standard Parameters

Parameters	**Description**	**Values**
β	Discount Factor (yearly)	0.96
γ	Risk Aversion (CRRA)	2
R	Interest Rate	1.02
σ	Death Probability	$\frac{1}{35}$
Z	Aggregate Conditions (normalized)	1
$\underline{\Phi}$	Borrowing Limit	0
b	Unemployment Benefit	0.1

I use standard values for the yearly discount factor, as well as for the risk aversion parameter in the CRRA utility function. The interest rate in my model grows by 2 percent per year.

I parametrize the stochastic probability of an agent dying at any period to 1/35, slightly less than 3 percent, to target an average worklife expectancy of 35 years with a geometric distribution. The aggregate economic conditions $\mathcal{Z}$ are normalized to one when there is no shock to output. I assign the borrowing limit to be zero to have a no-borrowing constraint. The borrowing constraint can be relaxed to allow a fixed level of borrowing $-\underline{\Phi}$ for all agents. The unemployment benefit value of 0.10 lends itself to be ten percent of the wage in the skill-insensitive (L) sector given the normalized parameter of $\mathcal{Z}$.

In table 2, I target the hiring and separation probabilities in both model sectors to match the standard separation rates seen in the data, 10 percent. The hiring probability in the low sector is assumed to be one, whereas the hiring probability in the high skill sensitive sector grows depending on the level of human capital. I follow Huckfeldt (2021) to calibrate the probabilities of human capital changes to wage growth.

TABLE 2. Hiring, Firing, and Human Capital Transition Parameters

Parameter	**Description**	**Value**
η_H	Separation Rate in High Sector	0.1
η_L	Separation Rate in Low Sector	0.1
$p_H(h)$	Prob of Hiring in High Sector by h	0.1 to 0.25
p_L	Prob of Hiring in Low Sector	1
π_H	Prob of human capital ↑ in High Sector	0.03
π_L	Prob of human capital ↓ in Low Sector	0.001
π_U	Prob of human capital ↓ in Unemployment	0.05

4. Counterfactual Exercises

I use the model outlined and parametrized in section 3 to run exercises that quantify the importance of the parental safety-net.

4.1. Policy Functions

To begin, Figure 5 below shows the sector policy functions given current assets a, human capital h, and current sector s. For illustrative purposes, the figure below shows an asset grid with fifteen levels of assets and five levels of human capital. A finer grid is used for the numerical calculations (see appendix). The asset level in Figure 5 is shown on the y-axis and the human capital level is denoted on the x-axis. The subplots represent the agent's current sector: from left to right, currently in unemployment (U), low skill-insensitive sector (L), and high (skill-sensitive) sector (H).

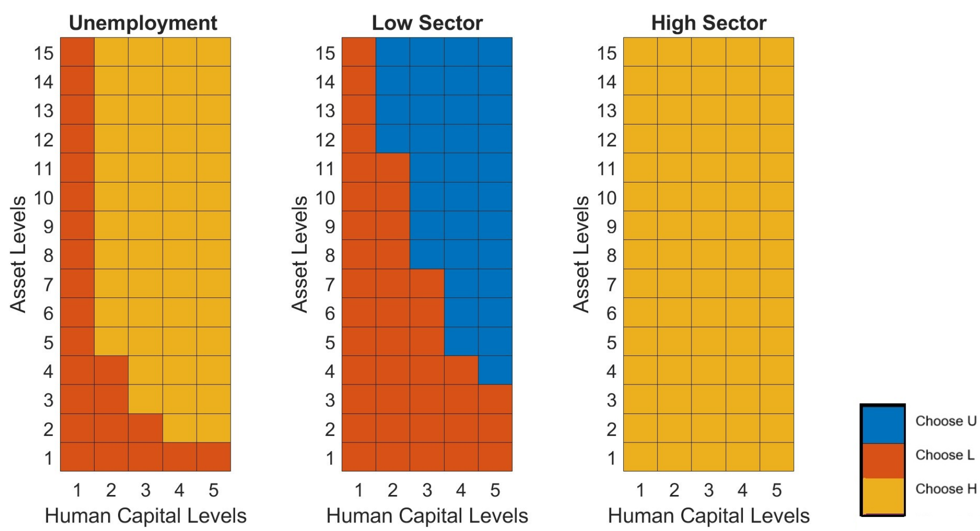

FIGURE 5. Sector Policy Functions

In Figure 5, the optimal policy choice for the sector next period is denoted by the following colors: blue indicates that the agent will optimally choose to enter unemployment, orange signifies that the agent will optimally choose to go into or stay in the skill-insensitive sector (L), while yellow represents the combinations of assets, human capital, and current sector where it is optimal to choose to target or remain in the skill-sensitive sector (H).

Starting with the left-most subplot for agents currently in unemployment, we see that the poorest individuals with very few assets, as well as all those with the lowest human capital level h equal to one, will optimally choose the low skill-insensitive sector next period in orange. For the latter group, with the lowest level of human capital $h = 1$, the earnings they would receive in the low skill-insensitive sector would be equivalent to those they would receive in the high sector where earnings depend on human capital, and in both cases greater than the unemployment benefit earned in unemployment. For values of human capital between zero and one, earnings would be even lower in the skill-complementary sector than in the low sector where all workers are paid the same. At the same time, more frictional hiring in the high sector makes targeting the high sector more risky, as agents are more likely to remain unemployed with reduced earnings for another period if they are not hired in the high sector. The low sector, in the current parametrization, hires with probability one, and even though potential earnings may be the same for those with the lowest level of human capital, agents will optimally choose to go into the low sector which has a higher hiring probability.

The former group, that has some human capital accumulated but very little assets, also chooses to enter the low sector. This is because individuals who are in unemployment do not accumulate assets, and the utility of being at the borrowing constraint is very low . Therefore, for individuals who find themselves at the borrowing constraint, with any level of human capital, will choose to enter the low sector next period in orange, rather than risk not being hired into the high sector, and staying in unemployment again next period.

The negative relationship between assets and human capital in preferring the low sector next period, in orange, comes from the continuation value of next period. Returns to human capital are large in the high sector, and therefore individuals with high human capital have a higher continuation value of trying the skill-sensitive sector even when they have very low assets, so long as they are not liquidity constrained. In Figure 5, individuals with high and very high human capital both have higher expected value in the high sector than in the low sector, and are willing to risk remaining in unemployment if not hired by the high sector up until they are at the borrowing constraint. In the skill-sensitive sector, the returns to human capital are comparatively lower for agents with medium to low human capital. Since the earnings difference between sectors is smaller for these individuals, and the hiring probability is lower for low human capital individuals to become employed in the high sector, individuals with less human capital require more assets to prefer the high skill-sensitive sector in yellow over the low skill insensitive sector in orange.

The middle subplot of Figure 5 visualizes next period's sector preferences for agents who are currently working in the low skill insensitive sector. As in the unemployment subplot, there is a negative relationship in assets and human capital between sector choice next period, namely preferring to remain in the low sector, in orange, and preferring to go into unemployment, in blue, to potentially be hired in the high sector. In the parametrization of the model, all individuals with the lowest human capital level prefer to remain in the low sector next period, regardless of their asset level. This is because for very low human capital types with human capital level one, their earnings will be equivalent in the low and high skilled sectors. These individuals will choose to remain employed in the low sector until they meet a separation shock or die. The returns to earnings of being employed in the high sector increase with human capital, and therefore as human capital increases, a smaller stock of assets is needed for individuals to prefer to leave the low skill insensitive sector.

For low asset holdings, all individuals, irrespective of their human capital level, prefer to remain in the low-sector in orange as opposed to risk unemployment to switch sectors. This asset threshold, however, is higher than the equivalent threshold for unemployed individuals and above the borrowing constraint. *This reveals a "trap"* . Pressed by liquidity constraints, unemployed workers with few assets prefer the low sector, which has a higher hiring probability but lower earnings potential. However these same individuals, once employed in the low sector, need to accumulate more assets for them to optimally want to switch sectors. Thus poor individuals, and especially poor individuals with human capital, who are near or at the borrowing constraint are more likely to be "stuck" in the low sector for several periods until they can save up to have the opportunity to switch to the high sector. The potential for human capital depreciation in the low sector makes it even more difficult for individuals to move sectors, as those with lower human capital require an even larger buffer of savings to make the risk of multiple periods unemployment worth the benefit of switching to the high sector. This phenomenon can significantly decrease their lifetime cumulative earnings compared to individuals born with more assets and richer parents, who choose the high sector directly from unemployment (and have more potential for human capital growth in the high sector as well).

The right-most subplot of Figure 5 shows that for agents currently working in the high sector, all agents would prefer to remain in that sector. These workers will continue to be employed in the high sector barring separation or death shocks.

4.2. Effect of Contraction of the aggregate economy

Figure 6 shows individuals next period sector preferences when the aggregate economy contracts. The simulated contraction is set to match the increase in the unemployment rate in Canada during the Great Recession (a contraction of 20%).

A contraction in aggregate economic conditions $\mathcal{Z}$ shrinks workers' earnings in both the low and high sectors, and decreases the hiring probability in the high sector for all levels of human capital. The policy function for individuals employed in the high sector does not change, and all high sector workers prefer to remain working in the high sector rather than going to unemployment or switching sectors.

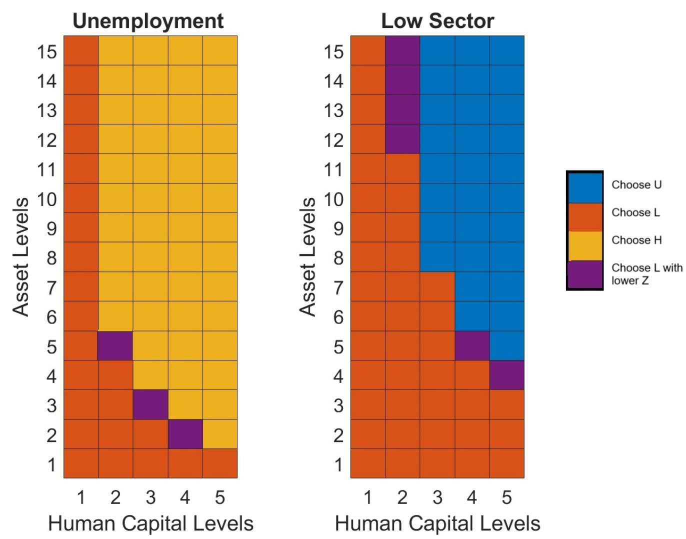

FIGURE 6. Sector Policy Functions when Aggregate Economy Contracts

However the contraction of the aggregate economy affects the choice of next period's sector depending on asset and human capital levels. In Figure 6, the purple demonstrates a different in sector choice depending on the aggregate labor market conditions. More specifically, purple highlights asset and human capital pairs where in worse economic conditions, individuals prefer to go to the low sector, but would have preferred the high sector in good aggregate economic conditions. These types of individuals are most affected by the worse state of the economy. We can consider these marginal individuals carefully when thinking of targets for policies aimed at reducing the scarring effect of recessions.

Individuals who are currently unemployed, in the furthest left subplot, require a larger stock of assets to prefer the high sector when the aggregate economy contracts. With the hiring probability in the high sector decreasing for all workers and the earnings premium of the high sector narrowing, the continuation value of trying for the high sector decreases during a recession for all but the highest human capital workers.

Workers in the skill-insensitive low sector, in the middle subplot, also need more assets to be willing to switch sectors compared to when the aggregate economy is stronger. The costs of the recession are especially high for workers with medium to low human capital, who for all levels of assets prefer to remain in the low sector. Thus, in a recession individuals need a larger stock of assets to optimally choose the high sector. For those with low assets who enter the low sector, they must accumulate even more assets to potentially switch sectors and for some workers with low human capital, it's no longer optimal to move sectors no matter how much they save.

The counterfactual effect of a recession on the distribution of workers across industry in the simulated economy is shown in Figure 7. The dark blue bars represent the share of workers across each sector in the baseline, when the aggregate economy is doing well. In the simulation, this is normalized with $\mathcal{Z}$ equal to one. The light blue bars demonstrate the share of workers by sector when there is a contraction to aggregate economy of 20%, corresponding to the increased growth in the unemployment rate during the height of the Great Recession.

In the simulated model, the share of workers employed in the low sector increases when aggregate economic conditions worsen, compared to the baseline. At the same time, fewer workers are employed in the high skill-sensitive sector under the simulated recession. This corresponds with the change in the policy functions shown in Figure 6. The simulated recession reduces the hiring probability p^H in the high skill-complementary sector for all levels of human capital, making it less likely for all unemployed workers to be hired in the high sector. At the same time, the shock to the aggregate economy lowers worker earnings in both sectors, with earnings in the high skill-complementary sector falling comparatively more than those in the low skill-insensitive sector. The resulting wage premium between the high and low sectors therefore shrinks under a simulated recession, which makes the continuation value in the low sector more appealing. In the simulation, a recession decreases the share of unemployed and non-employed workers compared to the baseline, which comes about from more workers preferring the low sector that is assumed to hire with probability one. Relaxing this assumption would better match the increase in unemployment observed in the data, and as long as the probability of being hired in the high sector is smaller than the hiring probability in the low sector, we would continue to observe a decrease in employment share in the high sector during a recession.

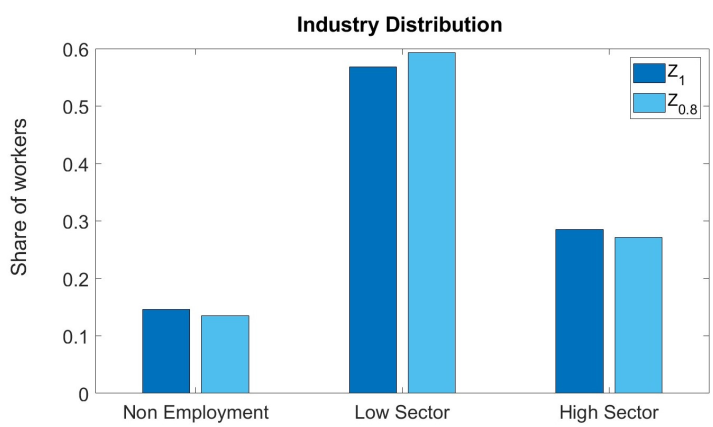

FIGURE 7. Sectoral Distribution when Aggregate Economy Contracts

4.3. Effect of Changing Initial Asset Distribution

To better understand the importance and policy implications of initial conditions on individual outcomes, I decompose the effect of initial assets and initial human capital, using the model to run counterfactual exercises.

First, I use the model to visualize the effect of changing only the initial asset distribution. This exercise is synonymous to changing the parental income distribution of new graduates. More specifically, I suppose that all individuals have the same initial asset level a_0, and allow initial human capital h_0 to vary. I compare the distribution in the baseline model with two counterfactual exercises where there is no income inequality, and all individuals are either born rich or poor, with high and low initial assets respectively.

Figure 8 demonstrates the share of workers in each sector in three scenarios. The baseline scenario is shown in dark blue. I display the counterfactual exercise where all individuals are born rich with high initial assets, shown in the figure in light blue, and the exercise where are individuals born poor with low initial assets in purple. I restrict only the initial assets, and allow for subsequent generations to inherit potentially unequal savings from their parents following a death shock.

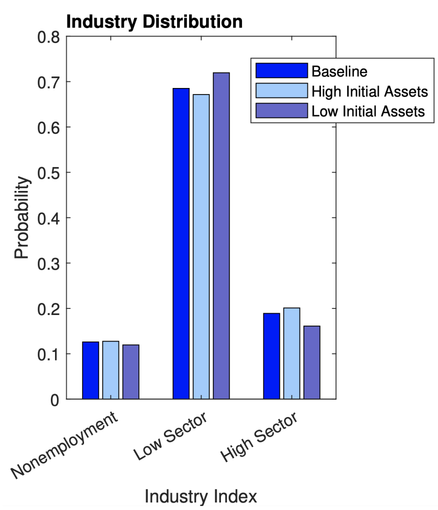

FIGURE 8. Sector Distribution by Initial Asset Level

In an economy where all individuals begin with high initial assets, Figure 8 reveals that there is a slightly larger share of workers employed in the high skill complementary sector compared to the baseline asset distribution. Similarly, there is a smaller proportion of workers working in the low sector when all individuals have high initial assets, and we can also see a slight increase in the proportion of non-employed workers compared to the baseline in Figure 8.

The intuition is as explained above. In partial equilibrium, workers who have larger initial assets have a larger parental safety net, and, all else equal, prefer to work in the skill-sensitive high sector. With a higher level of initial assets, individuals are more willing to risk additional periods of unemployment while waiting to be hired in the high sector, since they are further from the borrowing constraint and less likely to be liquidity constrained.

In comparison, in a world where all individuals begin poor with lower initial assets, shown in Figure 8 in purple, there are fewer workers employed in the high sector and more workers are employed in the low sector compared to the baseline scenario. In the model, non-employment decreases when all individuals have low initial assets because of the added proximity to the borrowing constraint. Individuals therefore prefer to be employed in the low sector with probability one and are less willing to risk prolonged unemployment to move to the high sector because of the increased probability of being liquidity constrained.

4.4. Effect of Changing Initial Human Capital Distribution

In the second counterfactual exercise, I decompose the impact of changing the initial human capital distribution, while keeping the initial asset distribution unchanged. To do so, I find the equilibrium distribution when individuals can have different initial assets, but all individuals start with the same initial human capital, h_0. This exercise speaks to the importance of ability disentangled from the direct effects of parental wealth.

Figure 9 illustrates the distribution of workers across sectors. The baseline scenario, shown in dark blue, is compared to the counterfactual exercise where individuals all begin with the same low level of human capital, displayed in light blue. I keep the initial asset distribution unchanged, and there are multiple levels of initial assets a_0 in this counterfactual exercise.

As Figure 9 demonstrates, when initial human capital is homogeneously low, the share of individuals working in the high sector decreases by approximately 15 percentage points compared to the baseline with human capital differences. The share of workers in the low sector also increases substantially from approximately 60% to 75% when all individuals begin with low initial human capital. Non-employment attenuates slightly as well from the baseline model, though the reduction is smaller in magnitude than the share of workers in the high sector.

Lowering initial human capital for all individuals reduces the payoff in the high sector, where earnings depend on human capital. Thus the trade-off shrinks between the skill-complementary high sector - where earnings depend on human capital - and skill-insensitive low sector, where earnings are independent of human capital. Recalling the

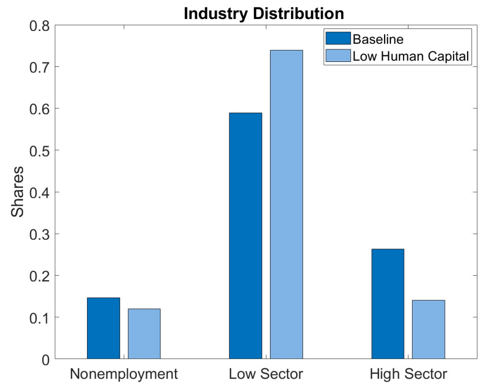

FIGURE 9. Sector Distribution by Initial Human Capital

policy functions in Figure 5, for given a level of assets, a currently unemployed individual is more likely to optimally prefer the high sector when their human capital is higher. Analogously, lower human capital is associated with preferring the low sector, holding assets and all else constant. Sector choice is especially sensitive to human capital when assets are low and individuals are closer to the borrowing constraint.

This explains the large increase in the share of workers employed in the low sector when initial human capital is homogeneously low compared to the baseline model. As more workers flock to the low sector, fewer workers optimally choose the high sector, and there are therefore fewer workers are hired in the high sector and there are fewer individuals in unemployment who preferred to join the high sector but were not hired in the current period. The differences in initial sector distribution are also amplified, as low sector workers have lower earnings potential and accumulate fewer assets than those in the high sector. This makes individuals who are separated from their jobs on average closer to the borrowing constraint when in unemployment, and in turn more likely to optimally select the low sector.

5. Conclusion

This paper studies the unequal impact of initial labor market conditions by parental income, and how it can create persistent differences in long-run outcomes. Using large Canadian administrative tax data, I analyze the relationship between parental income and child labor market trajectories, focusing on how aggregate economic conditions influence the wage dynamics of new labor market entrants. I find significant and lasting effects on wages that are unequal by parental income group, with entrants from lower-income families being more vulnerable to downturns and experiencing larger earnings declines when starting their careers during a recession. Interestingly, these disparities are largely mitigated when accounting for sector of employment, which suggests that the sector selection may play an important role in explaining these labor market dynamics. Motivated by these insights, I develop a model incorporating financial constraints and frictional sector choices within a heterogeneous agent model to quantify the role of initial parental income.

The mechanism I explore is parental income acts as a safety net, whereby individuals with poorer parents find themselves closer to the borrowing constraint compared to their peers with better-off parents. In settings where there are larger search frictions in the high-paying sectors, those with poorer parents are more likely to choose into jobs with low earnings and low human capital growth. When the initial labor market conditions worsen such as in a recession, this mechanism is amplified as the job finding rate decreases.

My findings help to better understand a key driver of inter generational inequality and have important policy implications. I suggest that targeted economic policy of unemployment insurance for new labor market entrants can alleviate scarring effects for marginal workers from poorer families. Policies of unemployment insurance for young workers entering the labor market can be especially effective when targeting individuals who, because of the recession, begin their careers in sectors with low earnings potential, but would have chosen otherwise with better initial labor market conditions. Such policies are particularly relevant in contexts where inter-generational income and human capital mobility is low.

A. Appendix

FIGURE 10. Unemployment Rate in Canada, 1975- 2020
Recessions shaded in gray

TABLE 3. Real Labor Earnings Response to 1 p.p. increase in initial UR_{c0}, bottom 20% and top 80% of parental income

	Dependent variable: T4E_cpi, Real Labor Earnings in 2002 CAD							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
UR Results j Years After Graduation								
Reference Group is Bottom 20% of Parental Income								
$\gamma_{0,j=0}$	-2,478.416***	-2,564.204***	-2,533.158***	-2,614.899***	-2,616.457***	-2,601.437***	-2,034.322***	-1,745.668***
	(269.735)	(266.568)	(264.923)	(262.212)	(262.215)	(256.766)	(256.87)	(234.447)
$\gamma_{0,j=1}$	-2,358.815***	-2,406.333***	-2,511.994***	-2,660.625***	-2,662.264***	-2,614.033***	-1,853.884***	-1,759.781***
	(273.03)	(269.825)	(268.16)	(265.415)	(265.417)	(259.896)	(257.385)	(237.307)
$\gamma_{0,j=2}$	-2,151.935***	-2,155.812***	-2,365.000***	-2,566.850***	-2,568.559***	-2,407.400***	-1,516.135***	-1,777.000***
	(273.512)	(270.301)	(268.636)	(265.886)	(265.889)	(260.355)	(260.971)	(237.723)
$\gamma_{0,j=3}$	-2,211.117***	-2,135.527***	-2,128.101***	-2,549.252***	-2,551.209***	-2,273.418***	-1,462.519***	-1,740.783***
	(338.807)	(334.83)	(332.763)	(329.363)	(329.366)	(322.506)	(322.494)	(294.467)
$\gamma_{0,j=4}$	-2,053.467***	-1,875.223***	-1,681.566***	-2,322.631***	-2,324.547***	-1,959.308***	-1,215.873***	-1,365.992***
	(392.456)	(387.85)	(385.458)	(381.53)	(381.533)	(373.588)	(372.416)	(341.104)
$\gamma_{0,j=5}$	-2,337.990***	-2,143.261***	-1,854.507***	-2,522.854***	-2,524.605***	-2,127.655***	-1,318.043***	-1,386.220***
	(399.288)	(394.602)	(392.171)	(388.177)	(388.179)	(380.096)	(380.492)	(347.047)
$\gamma_{0,j=6}$	-2,362.460***	-2,128.133***	-1,836.438***	-2,493.395***	-2,495.023***	-2,103.494***	-1,677.638***	-1,588.242***
	(419)	(414.083)	(411.532)	(407.337)	(407.339)	(398.856)	(399.58)	(364.175)
$\gamma_{0,j=7}$	-2,116.639***	-1,963.095***	-1,779.154***	-2,417.304***	-2,419.031***	-2,044.882***	-1,268.630***	-1,313.457***
	(448.224)	(442.963)	(440.23)	(435.738)	(435.74)	(426.665)	(426.542)	(389.568)
$\gamma_{0,j=8}$	-1,645.904***	-1,570.259***	-1,522.383***	-2,178.521***	-2,180.230***	-1,929.869***	-1,074.756**	-1,170.018***
	(468.911)	(463.406)	(460.545)	(455.843)	(455.845)	(446.349)	(447.174)	(407.54)
$\gamma_{0,j=9}$	-1,118.639**	-1,038.177**	-1,095.817**	-1,725.721***	-1,727.441***	-1,548.440***	-806.555*	-897.143**
	(481.692)	(476.037)	(473.099)	(468.265)	(468.267)	(458.512)	(458.828)	(418.646)
$\gamma_{0,j=10}$	175.217	243.156	56.43	-588.499	-590.29	-361.678	484.924	172.555
	(485.019)	(479.325)	(476.368)	(471.5)	(471.502)	(461.681)	(463.8)	(421.537)
$\gamma_{0,j=11}$	1,753.597***	1,880.162***	1,748.482***	929.989*	928.163*	1,178.628**	1,518.468***	1,255.467***
	(519.932)	(513.829)	(510.658)	(505.449)	(505.45)	(494.922)	(500.58)	(451.885)
UR x Family Income Group Results								
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income								
$\gamma_{p,j=0}$	490.642*	483.262*	540.715**	546.344**	545.698**	440.186*	215.165	134.38
	(277.879)	(274.617)	(272.922)	(270.128)	(270.128)	(264.503)	(264.125)	(241.499)
$\gamma_{p,j=1}$	910.644***	901.228***	695.978**	692.424**	691.784**	640.292**	320.319	362.914
	(284.312)	(280.975)	(279.244)	(276.384)	(276.385)	(270.628)	(267.373)	(247.092)
$\gamma_{p,j=2}$	1,010.040***	1,001.086***	828.098***	819.889***	819.264***	752.918***	262.824	545.436**
	(286.629)	(283.264)	(281.518)	(278.635)	(278.636)	(272.832)	(272.733)	(249.102)
$\gamma_{p,j=3}$	676.469*	717.560**	555.255	600.813*	600.115*	617.670*	337.132	660.376**
	(357.404)	(353.209)	(351.03)	(347.436)	(347.436)	(340.199)	(339.273)	(310.61)
$\gamma_{p,j=4}$	93.049	146.454	96.162	234.268	233.635	348.357	43.025	275.185
	(415.528)	(410.65)	(408.115)	(403.937)	(403.937)	(395.521)	(393.225)	(361.12)
$\gamma_{p,j=5}$	-23.355	28.508	-22.738	136.114	135.288	216.641	-231.831	-69.354
	(423.72)	(418.746)	(416.161)	(411.9)	(411.901)	(403.319)	(402.533)	(368.242)
$\gamma_{p,j=6}$	-244.258	-262.371	-314.141	-136.774	-137.694	-88.414	-267.632	-212.056
	(445.012)	(439.787)	(437.073)	(432.596)	(432.597)	(423.583)	(423.202)	(386.743)
$\gamma_{p,j=7}$	-598.269	-574.818	-601.232	-381.399	-382.177	-359.284	-960.321**	-729.337*
	(476.176)	(470.585)	(467.681)	(462.889)	(462.89)	(453.245)	(452.143)	(413.829)
$\gamma_{p,j=8}$	-609.917	-547.585	-567.683	-264.921	-265.829	-183.363	-819.423*	-556.609
	(498.199)	(492.351)	(489.311)	(484.3)	(484.3)	(474.21)	(474.032)	(432.969)
$\gamma_{p,j=9}$	-932.976*	-893.753*	-949.208*	-647.99	-648.883	-521.358	-1,212.286**	-916.468**
	(511.649)	(505.642)	(502.521)	(497.373)	(497.374)	(487.01)	(486.559)	(444.657)
$\gamma_{p,j=10}$	-1,274.259**	-1,222.922**	-1,261.372**	-945.286*	-946.083*	-846.190*	-1,658.908***	-1,160.722***
	(515.017)	(508.971)	(505.829)	(500.648)	(500.648)	(490.217)	(491.577)	(447.584)
Age	Y	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y	Y
Degree Type					Y	Y	Y	Y
College or Uni						Y	Y	Y
Major						Y	Y	Y
Industry FE Two-Digit NAICS							Y	
Two-Digit NAICS, where 0 is not employed								Y

TABLE 4. Real Log Positive Labor Earnings Response to 1 p.p. increase in initial UR_{c0}, bottom 20% and top 80% of parental income

	Dependent variable: log (T4E_cpi), Real Positive Log Labor Earnings							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
UR Results j Years After Graduation								
Reference Group is Bottom 20% of Parental Income								
$\gamma_{0,j=0}$	-0.057***	-0.058***	-0.057***	-0.060***	-0.060***	-0.061***	-0.053***	-0.054***
	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=1}$	-0.031***	-0.032***	-0.036***	-0.040***	-0.040***	-0.039***	-0.033***	-0.035***
	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=2}$	0.001	0.001	-0.007	-0.012	-0.012	-0.007	0.001	-0.003
	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=3}$	-0.025**	-0.022**	-0.020*	-0.031***	-0.031***	-0.022**	-0.012	-0.014
	(0.011)	(0.011)	(0.011)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
$\gamma_{0,j=4}$	-0.024*	-0.018	-0.01	-0.026**	-0.027**	-0.013	-0.007	-0.007
	(0.013)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.011)	(0.011)
$\gamma_{0,j=5}$	-0.034***	-0.028**	-0.016	-0.033***	-0.033***	-0.019	-0.009	-0.009
	(0.013)	(0.013)	(0.013)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
$\gamma_{0,j=6}$	-0.055***	-0.048***	-0.037***	-0.055***	-0.055***	-0.040***	-0.028**	-0.027**
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.012)	(0.012)
$\gamma_{0,j=7}$	-0.016	-0.012	-0.004	-0.022	-0.022	-0.01	0.001	0.001
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.013)	(0.013)
$\gamma_{0,j=8}$	-0.013	-0.011	-0.009	-0.028*	-0.028*	-0.02	-0.011	-0.01
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)
$\gamma_{0,j=9}$	0.009	0.012	0.008	-0.011	-0.012	-0.006	0.005	0.004
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
$\gamma_{0,j=10}$	0.029*	0.031**	0.024	0.004	0.004	0.01	0.018	0.019
	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
$\gamma_{0,j=11}$	0.068***	0.072***	0.067***	0.044***	0.044***	0.051***	0.062***	0.060***
	(0.017)	(0.017)	(0.017)	(0.016)	(0.016)	(0.016)	(0.015)	(0.015)
UR x Family Income Group Results								
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income								
$\gamma_{p,j=0}$	0.026***	0.025***	0.028***	0.027***	0.027***	0.022***	0.020**	0.021***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
$\gamma_{p,j=1}$	0.032***	0.031***	0.021**	0.019**	0.019**	0.017**	0.019**	0.021**
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
$\gamma_{p,j=2}$	0.020**	0.020**	0.011	0.01	0.009	0.006	0.007	0.01
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
$\gamma_{p,j=3}$	0.020*	0.021*	0.012	0.013	0.013	0.012	0.014	0.015
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.01)	(0.01)
$\gamma_{p,j=4}$	(0.016	(0.015	(0.018	(0.014	(0.014	(0.012	(0.006	(0.006
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.012)	(0.012)	(0.012)
$\gamma_{p,j=5}$	-0.018	-0.016	-0.019	-0.014	-0.014	-0.012	-0.01	-0.01
	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.012)	(0.012)
$\gamma_{p,j=6}$	0.0004	-0.001	-0.001	0.005	0.005	0.005	0.004	0.003
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.013)	(0.013)	(0.013)
$\gamma_{p,j=7}$	-0.032**	-0.031**	-0.032**	-0.025*	-0.025*	-0.024*	-0.024*	-0.024*
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)
$\gamma_{p,j=8}$	-0.024	-0.021	-0.022	-0.012	-0.012	-0.01	-0.008	-0.01
	(0.016)	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
$\gamma_{p,j=9}$	-0.036**	-0.035**	-0.037**	-0.026	-0.026	-0.021	-0.022	-0.021
	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)
$\gamma_{p,j=10}$	-0.026	-0.025	-0.028*	-0.017	-0.017	-0.013	-0.009	-0.01
	(0.017)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.015)	(0.015)
Age	Y	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y	Y
Degree Type					Y	Y	Y	Y
College or Uni						Y	Y	Y
Major						Y	Y	Y
Industry FE Two-Digit NAICS							Y	
Two-Digit NAICS, where 0 is not employed								Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331	0.33
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331	0.33

TABLE 5. Real Log Positive Full-Time Labor Earnings Response to 1 p.p. increase in initial UR_{c0}, bottom 20% and top 80% of parental income

	Dependent variable: log (T4E_cpi), Real Positive Log Labor Earnings above 10k							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
UR Results j Years After Graduation								
Reference Group is Bottom 20% of Parental Income								
$\gamma_{0,j=0}$	-0.034***	-0.034***	-0.034***	-0.035***	-0.036***	-0.035***	-0.032***	-0.032***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)
$\gamma_{0,j=1}$	-0.040***	-0.040***	-0.040***	-0.041***	-0.042***	-0.039***	-0.033***	-0.034***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{0,j=2}$	-0.039***	-0.038***	-0.040***	-0.044***	-0.045***	-0.040***	-0.030***	-0.032***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{0,j=3}$	-0.034***	-0.032***	-0.031***	-0.041***	-0.042***	-0.036***	-0.026***	-0.026***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{0,j=4}$	-0.028***	-0.023***	-0.021***	-0.037***	-0.037***	-0.029***	-0.021***	-0.021***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.006)	(0.006)
$\gamma_{0,j=5}$	-0.030***	-0.026***	-0.022***	-0.038***	-0.038***	-0.030***	-0.023***	-0.022***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{0,j=6}$	-0.035***	-0.030***	-0.027***	-0.044***	-0.044***	-0.035***	-0.025***	-0.025***
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{0,j=7}$	-0.027***	-0.023***	-0.021**	-0.037***	-0.037***	-0.029***	-0.022***	-0.022***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)
$\gamma_{0,j=8}$	-0.013	-0.011	-0.009	-0.028*	-0.028*	-0.02	-0.011	-0.01
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)
$\gamma_{0,j=9}$	-0.012	-0.008	-0.009	-0.026***	-0.026***	-0.021**	-0.012	-0.012
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=10}$	-0.003	-0.001	-0.003	-0.019**	-0.019**	-0.012	-0.003	-0.004
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=11}$	0.028***	0.031***	0.031***	0.009	0.009	0.016*	0.025***	0.025***
	(0.010)	(0.010)	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
UR x Family Income Group Results								
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income								
$\gamma_{p,j=0}$	0.012*	0.011*	0.012*	0.014**	0.014**	0.015**	0.017***	0.017***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{p,j=1}$	0.015***	0.016***	0.012**	0.012**	0.012**	0.012**	0.013***	0.014***
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{p,j=2}$	0.017***	0.016***	0.014***	0.014***	0.014***	0.014***	0.012**	0.014***
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{p,j=3}$	0.006	0.007	0.005	0.005	0.005	0.008	0.007	0.007
	(0.007)	(0.007)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{p,j=4}$	-0.007	-0.007	-0.007	-0.005	-0.005	-0.002	-0.001	-0.001
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{p,j=5}$	-0.007	-0.007	-0.007	-0.005	-0.005	-0.002	-0.001	-0.001
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{p,j=6}$	-0.003	-0.004	-0.004	0.001	0.001	0.002	0.0004	0.0004
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)
$\gamma_{p,j=7}$	-0.012	-0.012	-0.012	-0.006	-0.006	-0.005	-0.003	-0.003
	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{p,j=8}$	-0.021**	-0.020**	-0.020**	-0.013	-0.013	-0.01	-0.011	-0.011
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)
$\gamma_{p,j=9}$	-0.021**	-0.020**	-0.021**	-0.012	-0.012	-0.009	-0.01	-0.01
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
$\gamma_{p,j=10}$	-0.021**	-0.020**	-0.021**	-0.013	-0.013	-0.011	-0.011	-0.01
	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
Age	Y	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y	Y
Degree Type					Y	Y	Y	Y
College or Uni						Y	Y	Y
Major						Y	Y	Y
Industry FE Two-Digit NAICS							Y	
Two-Digit NAICS, where 0 is not employed								Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331	0.33
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331	0.33